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THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF COMPUTING

EXAMINATION

Course : Broad Discipline of COMP-61431, BSc IMT-42477

Subject : COMP4422 Computer Graphics

Group : 1011, 131

Session : 2018 / 2019 Semester I

Date : 10 December 2018 Time : 12:30 - 14:30

Time Allowed: 2 Hours Subject Lecturer: Prof. BACIU George

This question paper has 15 pages.

Instructions to Candidates:

No aids are permitted other than very simple calculators.
Please answer the following questions in the space provided.

Do not turn this page until you are told to do so!

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COMP 407/COMP 4422 – Computer Graphics
Fall 2018 – Final Examination
Department of Computing
The Hong Kong Polytechnic University

Program Code*	Family Name	Given Name	Student ID	Final Mark

Instructions: **No** aids are permitted other than very simple calculators. Please answer the following questions in the space provided.

Question	Total Mark	Given Marks
1	10	
2	10	
3	10	
4	20	
5	20	
6	10	
7	10	
8	10	
Total Marks	100	

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Question 1: Light and Visible Spectrum

(a) [5/100] Describe and illustrate the characteristics of light as a wave and show why we can describe light in terms of frequency or wavelength.

(b) [5/100] Describe the visible electromagnetic spectrum and show the relative frequencies/wavelengths within its given range. Which are the lowest and highest color bands in terms of wavelength?

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Question 2: GPU Architecture

(a) [5/100] Illustrate the GPU architecture with the complete data path and explain the logical CPU-GPU interface. Clearly show the data paths between the main computer and the GPU system and within the GPU system.

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(b) [5/100] Briefly describe the functions of (1) the vertex shader and (2) the fragment shader. Illustrate their connectivity.

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Question 3: Vertex Transformations

(a) [5/100] Illustrate with a diagram and give examples of webgl code instructions for the three major vertex transformations in the GPU pipeline. Indicate the coordinate type for the input and output data between each transformation stage.

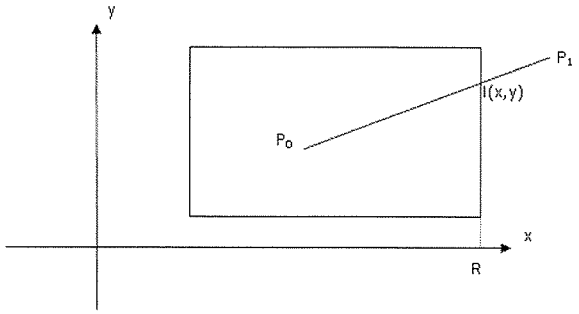
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(b) [5/100] For the window-to-viewport transformation show the steps in the formulation of the transformation matrix and explain why each step is necessary.

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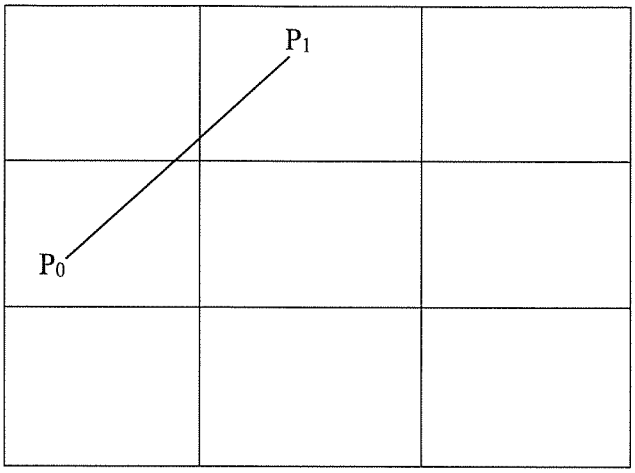
Question 4: Line Clipping

(a) [10/100] Consider the intersection between a line and the left window edge at x = R. Give the intersection point I(x,y) in terms of the line parameter *t* assuming the following case illustrated below:



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(b) [5/100] For the regions in the plane, please label each region with the codes for the 2D clipping algorithm discussed in class.



(c) [5/100] For the line P_0P_1 illustrated above, assuming that t_0 starts at P_0 and t_1 starts at P_1, what condition tells us that the line crosses outside the window?

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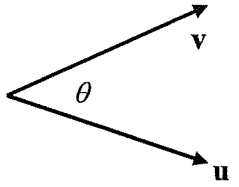
Question 5: Vectors and Transformations

(a) [5/100] From the following illustration, suggest the mathematical conditions for (i) the case when the ray enters the polygon, (ii) when the ray exists the polygon, and (iii) when the ray is parallel to one of the sides.

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(b) [5/100] The vector cross-product is used frequently in computing the normal vector on a surface at a given point. The geometric vector-cross product is given by:

$$\mathbf{u} \times \mathbf{v} = \mathbf{w}|\mathbf{u}||\mathbf{v}|\sin(\theta)$$



where θ is the angle between the two vector **u** and **v**.
Assume that **u** and **v** are unit vectors in the x-y plane and that we are using a right-handed coordinate system. Another way to obtain the cross-product is from the matrix operator given by:

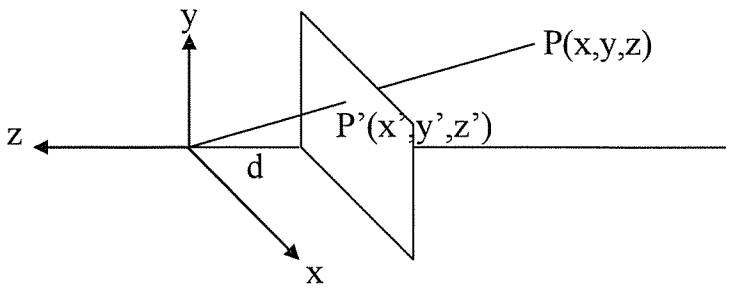
$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} 0 & -u_z & u_y \\ u_z & 0 & -u_x \\ -u_y & u_x & 0 \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$$

Derive the mathematical expression for the angle θ from the above information?

(c) [5/100] What is the length of the resulting vector and in which direction does it point?

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(d) [2/100] Given a point P(x,y,z) in space as shown below in the figure, derive the coordinates of the projection point P'(x',y',z') in the projection plane at z = -d.



(e) [3/100] From the derivation in (d) above, derive the homogeneous projection matrix that takes the homogeneous representation of point P(x,y,z) onto the homogeneous representation of the projected point P'(x',y',z').

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Question 6: Texture Mapping

(a) **[5/100]** Bilinear interpolation is often used in weighing the texture sample based on its distance from the pixel coordinates. Describe the bilinear interpolation using a diagram to illustrate the mapping and give the mapping formulation.

(b) **[5/100]** For MipMap texture mapping the image samples at different levels should be of what size and shape? Why?

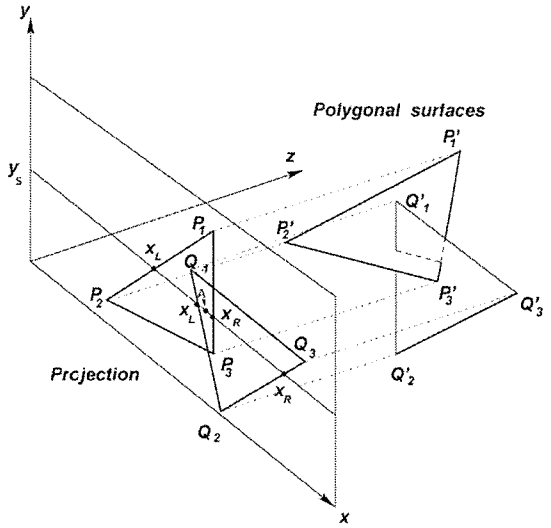
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Question 7: **Visibility**

(a) [5/100] A common way to render a list of objects is to choose an ordering such that the closest object to the eye is rendered last. What is this approach called and what is the problem with this approach? How to solve this problem?

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(b) [5/100] What is the significance of the following diagram and describe where and how it is used?



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Question 8: Rendering

[10/100] Draw a diagram and write down the irradiance equation. Explain the complete light reflection model and define all the terms in the equation. Finally, state what the equation represents.

END OF PAPER